

Software Requirements Specification (SRS)

Blood Bank & Donation Management System

Project: A System to Manage Blood Donation and Availability in Blood Banks

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1.0 — 06-09-2025 — Team	Final SRS draft for mini-project

Approvals

Role	Name / Signature / Email / Date
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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for a Blood Bank & Donation Management System. It defines the functional and non-functional requirements, external interfaces, and verification criteria for a platform that manages blood donors, blood inventory, donation drives, and hospital requests. It is intended as the baseline reference for the team's GitHub organisation, Jira epics and mini-project design.

1.2 Scope

The system covers: donor registration and eligibility screening; blood unit collection, testing and inventory tracking with expiry management; donation drive/camp scheduling and slot booking; hospital blood request submission, cross-matching and dispatch; and role-based access for Donors, Hospital Representatives, Blood Bank Staff/Lab Technicians, and Administrators. It excludes the internal clinical/laboratory testing equipment logic beyond recording results, and excludes hospital-internal patient record systems beyond the request/dispatch interface.

1.3 Audience

Developers, Testers, Hospital IT Technicians, blood bank staff, hospital representatives and donors who represent the system's end users.

1.4 Definitions, Acronyms & Abbreviations

- FR / NFR — Functional Requirement / Non-Functional Requirement
- RBAC — Role-Based Access Control
- OTP — One-Time Password
- PRBC — Packed Red Blood Cells
- FFP — Fresh Frozen Plasma
- QR — Quick Response (code)
- RTM — Requirements Traceability Matrix
- TLS — Transport Layer Security
- WCAG — Web Content Accessibility Guidelines

2. Overall description

2.1 Product perspective

The Blood Bank & Donation Management System is a simple web application that connects donors, blood bank staff and hospitals. It replaces manual registers/spreadsheets used to track donor eligibility, stock and hospital requests.

2.2 Major product functions

- Donor registration, identity verification and eligibility screening
- Blood unit collection, barcode/QR tagging, component classification and expiry tracking
- Real-time inventory search and dashboard
- Hospital blood request submission, cross-matching, approval and dispatch
- Donation drive/camp scheduling and donor slot booking
- Automated SMS/email notifications and reminders
- Administrative reporting, audit logging and role management

2.3 User roles and characteristics

- Donor: general public; expects a simple mobile-friendly flow to register, check eligibility and book slots.
- Hospital Representative: needs fast, reliable emergency request submission with clear status tracking.
- Blood Bank Staff / Lab Technician: manages day-to-day inventory, testing results and dispatch; needs an efficient desktop dashboard.
- Drive Coordinator: NGO/blood-bank staff who schedule and manage donation camps.
- System Administrator: manages accounts, roles, and views audit/compliance reports.

2.4 Operating environment

Web application, usable on a normal browser (Chrome/Firefox/Edge) on desktop or mobile, with a database on the backend.

2.5 Constraints

- Donor/patient data should be handled carefully, not exposed publicly.
- Blood group compatibility must follow standard rules (e.g. O- is a universal donor).

3. External interface requirements

3.1 User interfaces

Simple web pages per role: Donor page (register, check eligibility, book slot, view history), Hospital page (submit/track request), Staff/Admin page (manage inventory, requests, drives).

3.2 Hardware interfaces

- None

3.3 Software interfaces

- A notification API/service (SMS or email) for OTPs and reminders.
- A database (e.g. MySQL/PostgreSQL/SQLite) for donor, inventory and request records.

3.4 Communications

- HTTPS between the front end and backend.
- Simple REST APIs (JSON) between front end, backend and the notification service.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow BBS-F-###. A total of 15 functional requirements are specified across five feature areas.

4.1 User Authentication & Role-Based Access Control

Description: Authenticate all users and restrict actions to those permitted for their role.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBS-F-001	The system shall support login for four roles: Donor, Hospital Rep, Blood Bank Staff, and Admin.	Functional	High	All users	AC-001: Each role sees only its own dashboard. Test: TC-Auth-01	Login credentials stored in database
BBS-F-002	The system shall verify a new user's identity using an OTP sent to their phone/email during signup.	Functional	High	Security	AC-002: Signup completes only after correct OTP entry. Test: TC-Auth-02	Notification API
BBS-F-003	The system shall lock an account for 30 minutes after 5 wrong password attempts in a row.	Functional	Medium	Security	AC-003: 6th wrong attempt shows 'account locked' message. Test: TC-Auth-03	Depends on login attempt counter

4.2 Donor Registration & Eligibility Screening

Description: Capture donor data, enforce medical eligibility rules, and maintain donation history.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBS-F-010	The system shall let a donor create a profile with blood group, date of birth, weight and contact details.	Functional	High	Donor	AC-010: Profile save fails if a required field is empty. Test: TC-Donor-01	Stored in database
BBS-F-011	The system shall check donor eligibility using simple rules: weight $\geq$ 45 kg, age 18-65, and at least 56 days since the last donation.	Functional	High	Blood Bank Staff	AC-011: A donor who donated 30 days ago is shown as not yet eligible. Test: TC-Donor-02	Depends on last donation date in DB
BBS-F-012	The system shall save each completed donation to the donor's history (date, blood bank, quantity).	Functional	High	Lab Technician	AC-012: New donation appears in the donor's history list. Test: TC-Donor-03	Stored in Database

4.3 Blood Inventory & Component Tracking

Description: Track every blood unit from collection through testing, storage, expiry and disposal.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBS-F-020	The system shall assign a unique ID to every blood bag when it is collected.	Functional	High	Lab Technician	AC-020: Each bag ID is searchable and shows its details. Test: TC-Inv-01	Auto-generated ID, stored in DB
BBS-F-021	The system shall record the component type (Whole Blood, RBC, Platelets, Plasma) and set	Functional	High	Lab Technician	AC-021: Expiry date is auto-filled once component type is chosen. Test: TC-Inv-02	Expiry rules hardcoded per component

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	an expiry date for each unit based on that type.					
BBS-F-022	The system shall mark a unit as expired and remove it from available stock once its expiry date passes.	Functional	High	Blood Bank Staff	AC-022: An expired unit no longer shows up in the available-stock list. Test: TC-Inv-03	Runs as a check on save/query, no scheduler needed

4.4 Hospital Requests & Blood Dispatch

Description: Allow hospitals to request blood and the system to match, approve and dispatch compatible units.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBS-F-030	The system shall let a hospital rep submit a blood request with blood group, component and quantity needed.	Functional	High	Hospital Rep	AC-030: Submitted request appears in the staff queue. Test: TC-Hosp-01	Stored in database
BBS-F-031	The system shall check available stock for compatible blood groups and suggest matching units for a request.	Functional	High	Blood Bank Staff	AC-031: An A+ request never suggests a B- unit. Test: TC-Hosp-02	Uses blood group compatibility table
BBS-F-032	The system shall let staff approve a request and mark the matched unit as dispatched.	Functional	Medium	Blood Bank Staff	AC-032: Approved request status changes to 'Dispatched'. Test: TC-Hosp-03	Updates request status in DB

4.5 Donation Camp & Drive Management

Description: Allow blood banks/NGOs to schedule drives and donors to discover and book slots.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBS-F-040	The system shall let a coordinator create a donation drive with a location, date and number of slots.	Functional	Medium	Drive Coordinator	AC-040: A drive cannot be created without at least one slot. Test: TC-Drive-01	Stored in database
BBS-F-041	The system shall let a donor view upcoming drives and book an open slot.	Functional	Medium	Donor	AC-041: A full slot can no longer be booked. Test: TC-Drive-02	Depends on drive/slot data
BBS-F-042	The system shall send a reminder notification to a donor before their booked slot.	Functional	Low	Donor	AC-042: Reminder is sent before the slot time. Test: TC-Drive-03	Depends on Notification API

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs follow BBS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
BBS-NF-001	Inventory search results shall load in under 1-2 seconds under normal use.	Performance	High	Search returns results quickly in manual testing. Test: TC-Perf-01
BBS-NF-002	The system should stay available during normal usage hours, with minimal downtime.	Availability	Medium	No unplanned downtime observed during demo/testing period.
BBS-NF-003	Donor passwords and personal details shall be stored securely, not in plain text.	Security / Privacy	High	DB check confirms passwords are hashed. Test: TC-Sec-01
BBS-NF-004	The UI shall be simple enough for a non-technical user (donor or hospital staff) to use without training.	Usability	Medium	A test user can complete registration/request without help. Test: TC-UX-01
BBS-NF-005	The system shall prevent two staff members from allocating the same blood unit to two different requests at the same time.	Data Integrity	High	Only one of two simultaneous allocation attempts succeeds. Test: TC-Conc-01

5.1 Security

5.1.1 Security Objectives

- Protect donor personal and medical data from being accessed by unauthorized users.
- Make sure only the right role (Donor, Hospital Rep, Staff, Admin) can perform actions meant for that role.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
BBS-SR-001	All login traffic shall use HTTPS.	Security	High	Site is only reachable over HTTPS. Test: TC-Sec-01
BBS-SR-002	Passwords shall be stored as hashed values, never in plain text.	Security	High	DB inspection shows hashed passwords. Test: TC-Sec-02
BBS-SR-003	A Donor account shall not be able to access Staff/Admin-only pages or actions.	Security	High	Donor login blocked from staff pages. Test: TC-Sec-03
BBS-SR-004	Key actions (approving a request, marking a unit expired) shall be logged with user and timestamp.	Security	Medium	Log entry created for each such action. Test: TC-Sec-04
BBS-SR-005	A user session shall expire after a period of inactivity and require re-login.	Security	Medium	Session times out after inactivity. Test: TC-Sec-05

6. Quality attributes & Acceptance tests

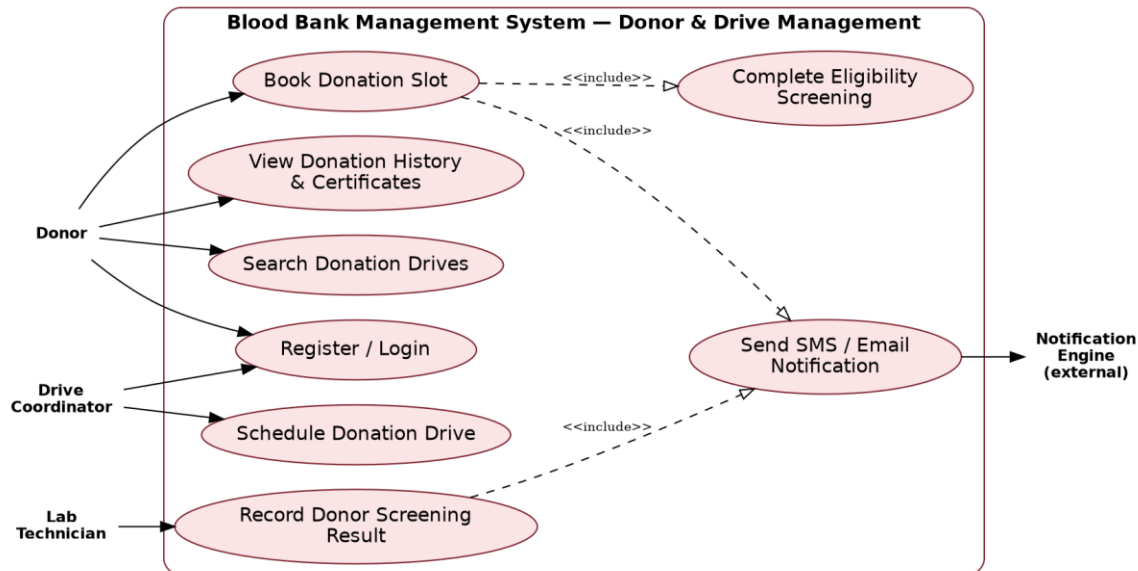
Exit criteria for acceptance: all High-priority functional requirements implemented and verified, no critical NFR failures, and the RTM shows all associated test cases passed.

Acceptance test suites: Authentication & RBAC, Donor Eligibility, Inventory & Expiry, Hospital Request & Cross-Matching, Drive Scheduling, Performance, Security, and Accessibility.

7. System models and diagrams

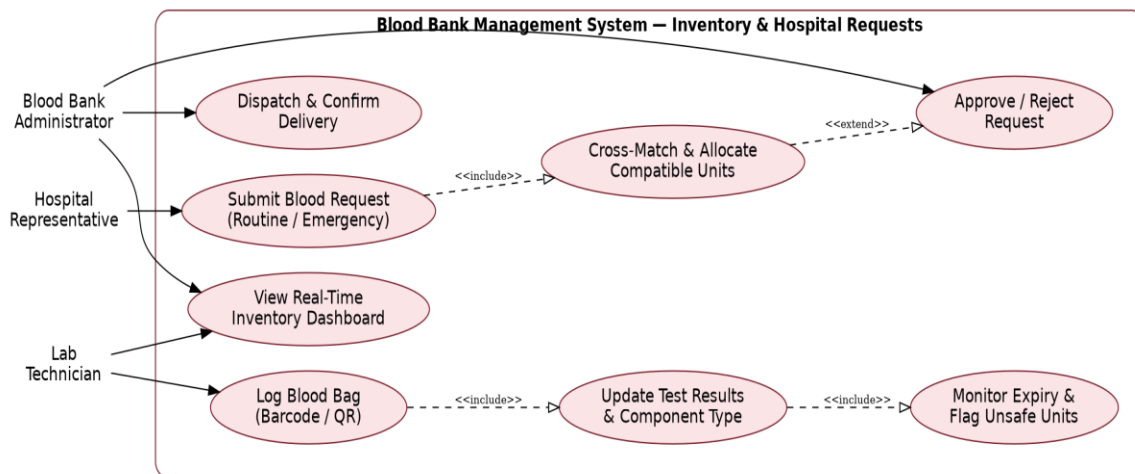
7.1 UML Use-Case Diagram — Donor & Drive Management

Covers registration/login, eligibility screening, drive discovery/booking, donation history, drive scheduling and donor-screening result recording.



7.2 UML Use-Case Diagram — Inventory & Hospital Requests

Covers blood bag logging, test-result entry, expiry monitoring, hospital request submission, cross-matching/allocation, approval and dispatch.



8. Requirements Traceability Matrix (RTM)

Representative sample; extend this matrix to cover every requirement ID once design specs and test cases are finalised.

Req ID	Requirement (short)	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
BBS-F-001	Role-based login	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N	
BBS-F-011	Eligibility engine	4.2 / DS-Donor-02	DonorModule	TC-Donor-02	N	
BBS-F-020	Barcode tracking	4.3 / DS-Inv-01	InventoryModule	TC-Inv-01	N	
BBS-F-023	Auto-dispose expired units	4.3 / DS-Inv-04	InventoryModule	TC-Inv-04	N	
BBS-F-031	Cross-match & allocate	4.4 / DS-Hosp-02	RequestModule	TC-Hosp-02	N	
BBS-F-041	Donor slot booking	4.5 / DS-Drive-02	DriveModule	TC-Drive-02	N	
BBS-NF-001	Search response time	5 / DS-Perf-01	InventoryModule / API	TC-Perf-01	N	
BBS-NF-006	No double allocation	5 / DS-Conc-01	RequestModule	TC-Conc-01	N	
BBS-SR-003	API-level RBAC	5.1.2 / DS-Sec-03	AuthModule / API Gateway	TC-Sec-03	N	